4 Downstream Evaluation

We assess whether profile ratings correlate with downstream recommendation performance. Investigating if higher-rated NL profiles also yield better recommendations, essential for ensuring both transparency and utility. We highlight that the goal is not to optimize recommendation performance, but rather to analyze its relation with user evaluations.

Following prior work, we adopt a recommendation approach by embedding NL profiles and item metadata into a shared latent space [10, 19]. Profiles are used as-is, while each track has associated metadata textual representation, as used in the prompts. Given both items and users textual representations, we fine-tune the bi-encoder proposed in [8] on our dataset by using their music-specific ranker as a teacher[4].

We use all the generated profiles for training. For each profile, the sampled tracks for each user–time window pair serve as positive examples, while negative examples are sampled from tracks the user has not interacted with in the year considered. In total, the training set comprised 100K user-positive-negative triplets. We train the model for 1 epoch using an AdamW optimizer, a batch size of 10, and a maximum sequence length fixed to 512. To test the model, for each user, we sampled 10 items the user has interacted with (but not used for profile generation) and 1K negative items, all of which not seen in the training set with that specific user. Our analysis is based on recall@10 and ndcg@10 metrics from the model's similarity scores, averaged over 5 runs with different random seeds. The code for implementation is made public in our repository[5].

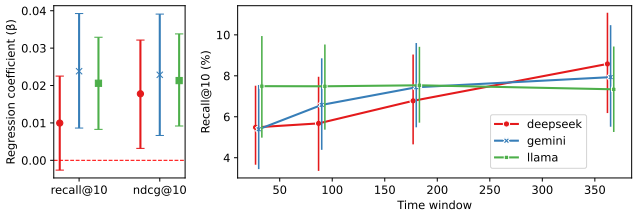

Figure 5: Left: Bootstrapped linear coefficients predicting recommendation scores from profile quality. Right: Mean Recall@10 with 95% CI for each model across time windows.

The left side of Figure 5 illustrates how $\delta r_{u,i}$ relates to recall@10 and ndcg@10, obtained by bootstrapping linear regression models that predict the normalized metrics from the user ratings, representing how much recommendation performance improves, on average, with a one-unit increase in user ratings. Except for Deepseek, all other metrics and models show a weak positive linear relationship. We provide more details on how recommendation performance relates to user and item characteristics in the Appendix. On the right side of Figure 5, we show the mean recall@10 with 95% CI for the different models and time windows. We note that the pattern is quite different from that in Figure 2. Except for Llama, the recommendation performance seems to improve with the longer time window, which can partially be explained by the fact that the test items were sampled from the entire consumption year.

[4]The ranker is implemented as a cross-encoders that take as input a music description concatenated with a set of music tags and produces as output a similarity score.
[5]https://github.com/deezer/recsys25_llm_biases

5 Discussion and Conclusion

This paper assessed the quality of LLM-generated natural language user profiles along two dimensions: user ratings and downstream recommendation performance. We compared profiles generated from items sampled over varying time windows of users' listening histories and found that the time window had only a minor impact on ratings. As long as key preferences were captured, users generally recognized themselves in the profiles. While some noted missing aspects of their taste (see Appendix), the largest variation in ratings stemmed from the choice of LLM rather than the data window. This variation likely reflects stylistic differences in the generated profiles: Gemini, for instance, produced shorter, high-level descriptions; Llama often emphasized specific artists and track names (despite prompt instructions), which some users preferred. DeepSeek occasionally highlighted metadata such as "remastered" in track titles, which users found less meaningful. Yet given the diversity in user preferences and model-specific biases, it would require a deeper qualitative analysis to expose what specific aspects were preferred.

We further examined how user and item characteristics influenced profile quality. Profiles of specialist users (those with high GS-scores) tended to receive higher ratings, suggesting their tastes may be easier to capture in natural language. Item characteristics, such as genre and country ratios, were also correlated with ratings, both positively and negatively. To account for potential confounds such as taste alignment, we employed a DR framework. The resulting ATE values confirmed that some content types systematically influenced ratings: for instance, rap content tended to reduce scores, while metal increased them, within the limits of our small user sample. Although it remains unclear whether these effects stem from metadata quality or model behavior, the fact that different LLMs responded differently to the same metadata suggests model-specific biases in content representation. These findings raise fairness concerns, as some users may consistently receive more representative profiles than others. Given that LLMs can inherit biases from the textual data on which they are trained, communities that are under-represented in digital corpora—particularly non-English-speaking ones, may be disadvantaged. We note, however, that our study included only three LLMs, and conclusions may not generalize broadly.

Moreover, user ratings and recommendation performance were only weakly correlated, revealing a misalignment between perceived representativeness and algorithmic utility. This is critical: if users do not recognize themselves in recommendation-optimized profiles, they may lose trust in the system. One potential solution is to decouple profile generation for user-facing experiences from those optimized solely for recommendation.

Future work should explore targeted fine-tuning or debiasing strategies to reduce representation gaps. In production systems, LLMs could be refined using feedback from user-corrected profiles, aiming to better balance personalization and fairness.

References

[1] Levin Brinkmann, Fabian Baumann, Jean-François Bonnefon, Maxime Derex, Thomas F Müller, Anne-Marie Nussberger, Agnieszka Czaplicka, Alberto Acerbi, Thomas L Griffiths, Joseph Henrich, et al. 2023. Machine culture. *Nature Human Behaviour* 7, 11 (2023), 1855–1868.

[2] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. 2022. Quantifying memorization across neural language models. *arXiv preprint arXiv:2202.07646* (2022).

[3] Jiawei Chen, Hande Dong, Xiang Wang, Fuli Feng, Meng Wang, and Xiangnan He. 2023. Bias and debias in recommender system: A survey and future directions. *ACM Transactions on Information Systems* 41, 3 (2023), 1–39.

[4] Google Cloud. 2025. Gemini 2.0 Flash: Generative AI on Vertex AI. https://cloud.google.com/vertex-ai/generative-ai/docs/models/gemini/2-0-flash. Accessed: 2025-04-30.

[5] Sunhao Dai, Chen Xu, Shicheng Xu, Liang Pang, Zhenhua Dong, and Jun Xu. 2024. Bias and unfairness in information retrieval systems: New challenges in the llm era. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 6437–6447.

[6] Yashar Deldjoo. 2024. Understanding biases in chatgpt-based recommender systems: Provider fairness, temporal stability, and recency. *ACM Transactions on Recommender Systems* (2024).

[7] Dario Di Palma, Felice Antonio Merra, Maurizio Sfilio, Vito Walter Anelli, Fedelucio Narducci, and Tommaso Di Noia. 2025. Do llms memorize recommendation datasets? a preliminary study on movielens-1m. *arXiv preprint arXiv:2505.10212* (2025).

[8] Elena V. Epure, Gabriel Meseguer-Brocal, Darius Afchar, and Romain Hennequin. 2024. Harnessing High-Level Song Descriptors towards Natural Language-Based Music Recommendation. In *Proceedings of the 3rd Workshop on NLP for Music and Audio (NLP4MusA2024)*. Association for Computational Linguistics.

[9] Michele Jonsson Funk, Daniel Westreich, Chris Wiesen, Til Stürmer, M Alan Brookhart, and Marie Davidian. 2011. Doubly robust estimation of causal effects. *American journal of epidemiology* 173, 7 (2011), 761–767.

[10] Zhaolin Gao, Joyce Zhou, Yijia Dai, and Thorsten Joachims. 2024. End-to-end Training for Recommendation with Language-based User Profiles. *arXiv preprint arXiv:2410.18870* (2024).

[11] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948* (2025).

[12] Jin Huang, Harrie Oosterhuis, Masoud Mansoury, Herke Van Hoof, and Maarten de Rijke. 2024. Going beyond popularity and positivity bias: Correcting for multifactorial bias in recommender systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 416–426.

[13] Gawesh Jawaheer, Peter Weller, and Patty Kostkova. 2014. Modeling user preferences in recommender systems: A classification framework for explicit and implicit user feedback. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 4, 2 (2014), 1–26.

[14] Chumeng Jiang, Jiayin Wang, Weizhi Ma, Charles LA Clarke, Shuai Wang, Chuhan Wu, and Min Zhang. 2024. Beyond Utility: Evaluating LLM as Recommender. *arXiv preprint arXiv:2411.00331* (2024).

[15] Kristina Matrosova, Lilian Marey, Guillaume Salha-Galvan, Thomas Louail, Olivier Bodini, and Manuel Moussallam. 2024. Do Recommender Systems Promote Local Music? A Reproducibility Study Using Music Streaming Data. In *Proceedings of the 18th ACM Conference on Recommender Systems*. 148–157.

[16] Daniel Müllensiefen, Bruno Gingras, Jason Musil, and Lauren Stewart. 2014. The musicality of non-musicians: An index for assessing musical sophistication in the general population. *PloS one* 9, 2 (2014), e89642.

[17] Ollama. 2025. Ollama: Open-Source Large Language Models. https://github.com/ollama/. Accessed: 2025-04-30.

[18] Pantelis Pipergias Analytis and Philipp Hager. 2023. Collaborative filtering algorithms are prone to mainstream-taste bias. In *Proceedings of the 17th ACM Conference on Recommender Systems*. 750–756.

[19] Filip Radlinski, Krisztian Balog, Fernando Diaz, Lucas Dixon, and Ben Wedin. 2022. On natural language user profiles for transparent and scrutable recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2863–2874.

[20] Jerome Ramos, Hossein A Rahmani, Xi Wang, Xiao Fu, and Aldo Lipani. 2024. Transparent and Scrutable Recommendations Using Natural Language User Profiles. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 13971–13984.

[21] Markus Schedl. 2019. Deep learning in music recommendation systems. *Frontiers in Applied Mathematics and Statistics* 5 (2019), 457883.

[22] Bruno Sguerra, Viet-Anh Tran, Romain Hennequin, and Manuel Moussallam. 2025. Uncertainty in Repeated Implicit Feedback as a Measure of Reliability. In *Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization*.

[23] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971* (2023).

[24] Viet-Anh Tran, Guillaume Salha-Galvan, Bruno Sguerra, and Romain Hennequin. 2024. Transformers Meet ACT-R: Repeat-Aware and Sequential Listening Session Recommendation. In *Proceedings of the 18th ACM Conference on Recommender Systems*. 486–496.

[25] Robin Ungruh, Karlijn Dinnissen, Anja Volk, Maria Soledad Pera, and Hanna Hauptmann. 2024. Putting Popularity Bias Mitigation to the Test: A User-Centric Evaluation in Music Recommenders. In *Proceedings of the 18th ACM Conference on Recommender Systems*. 169–178.

[26] Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. 2023. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926* (2023).

[27] Lanling Xu, Junjie Zhang, Bingqian Li, Jinpeng Wang, Mingchen Cai, Wayne Xin Zhao, and Ji-Rong Wen. 2024. Prompting large language models for recommender systems: A comprehensive framework and empirical analysis. *arXiv preprint arXiv:2401.04997* (2024).

[28] Joyce Zhou, Yijia Dai, and Thorsten Joachims. 2024. Language-Based User Profiles for Recommendation. *arXiv preprint arXiv:2402.15623* (2024).

Appendix

A Complementary Data Analysis

This appendix provides supplementary data explorations that complement the main paper.

Figure 6 shows the distribution of the considered users characteristics of the participants. Figure 7 complement the explorations of Section 3.1. Here, the evaluated characteristics are binned into 3 linear bins and we plot the mean $\delta r_{u,i}$ with a 95% CI for each bin. This serves for illustrating the relationship of the user characteristics and the ratings.

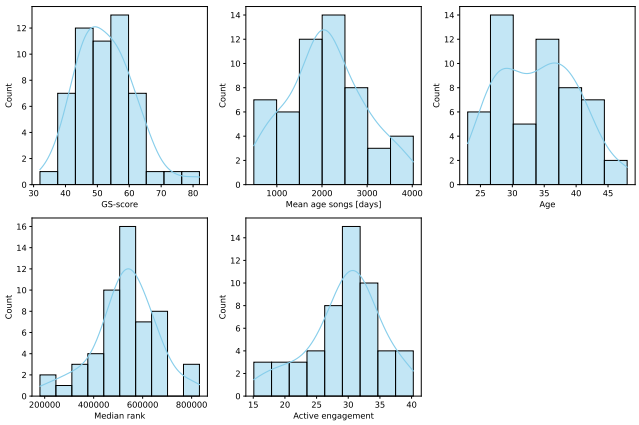

Figure 6: Distribution of the user characteristics.

Figure 8 and Figure 9 illustrate the bootstrapped mean β and 95% CI from fitting 5K linear regression predicting Recall@10 and NDCG@10 from each considered user characteristics (normalized). Here we note that, contrary to the Figure 3, the median rank, or the popularity of the tracks listened by the user, correlated positively with the metrics, meaning that users with preference for popular tracks have higher recommendation scores.

Figures 10 and 11 present the bootstrapped mean β coefficients and 95% CI from 5K linear regressions predicting Recall@10 and NDCG@10 based on the sampled item characteristics. We exclude associations with CI intervals crossing zero or with width greater

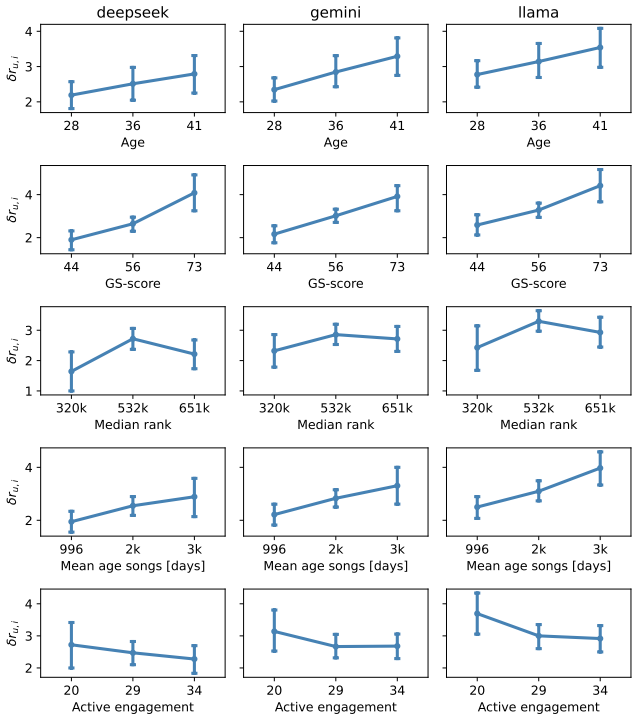

Figure 7: Mean $\delta r_{u,i}$ and 95% CI for user characteristics linearly binned into 3 groups.

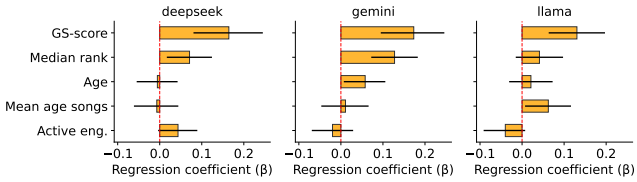

Figure 8: Bootstrapped estimated linear coefficients of user characteristics predicting recall@10.

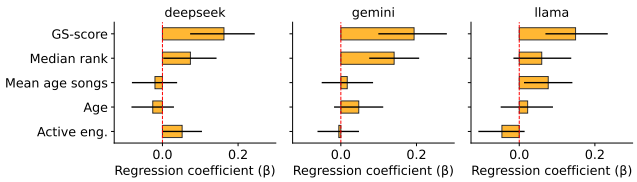

Figure 9: Bootstrapped estimated linear coefficients of user characteristics predicting ndcg@10.

than 10, as these indicate high uncertainty and lack of a reliable effect.

B Full Prompt and NL profile

For generating the profiles we used the same prompt format with a different list of items. All prompts started with the same preamble, followed by the item metadata and finishing with "Only return the textual summary — no explanation, no lists." We do not include profile examples in the prompt, allowing for flexibility.

In this section, we show different model generations given the following input prompt, with items collected from a 90 day time-window:

Here is a list of items consumed by a single user. From this list, generate a short and scrutable summary of their musical taste for music recommendation purposes. The summary should be a single, coherent, and informative paragraph that clearly conveys the user's preferences in a natural, human-readable way.

Base the description on the following list of consumed items and their metadata, without focusing on the names of tracks and artists. The items are presented in descending order:

title: This is a Mood
playcount: 19
artist: Franc Moody
origin country: United Kingdom
album release date: 2020
main genre: Electronic music
secondary genre: Nu-disco, Synthpop, Electronica

title: CAFé CON RON
playcount: 11
artist: Bad Bunny
origin country: Puerto Rico
album release date: 2025
main genre: Rap music
secondary genre: Caribbean music, Dancehall, Dance rap, Latin hip hop, Latin

title: Te Felicito (Turreo Edit)
playcount: 9
artist: Shakira
origin country: Colombia
album release date: 2023
main genre: Pop music
secondary genre: Latin trap, Pop rock, Electro pop, Contemporary R&B, Reggaeton, Latin american music

title: Y sólo se me ocurre amarte (Unplugged)
playcount: 9
artist: Alejandro Sanz
origin country: Spain
album release date: 2004
main genre: Pop music
secondary genre: Pop rock, Latin

title: Feel So Close (As Made Famous By Calvin Harris Instrumental Cover)
playcount: 9
artist: Calvin Harris
origin country: United Kingdom
album release date: 2012

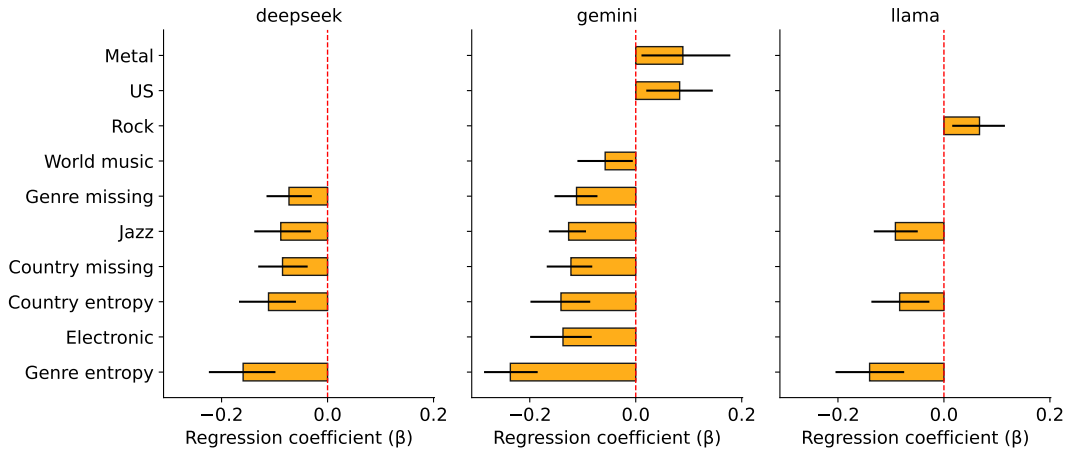

Figure 10: 5K bootstrapped estimated linear coefficients of sampled item characteristics predicting recall@10.

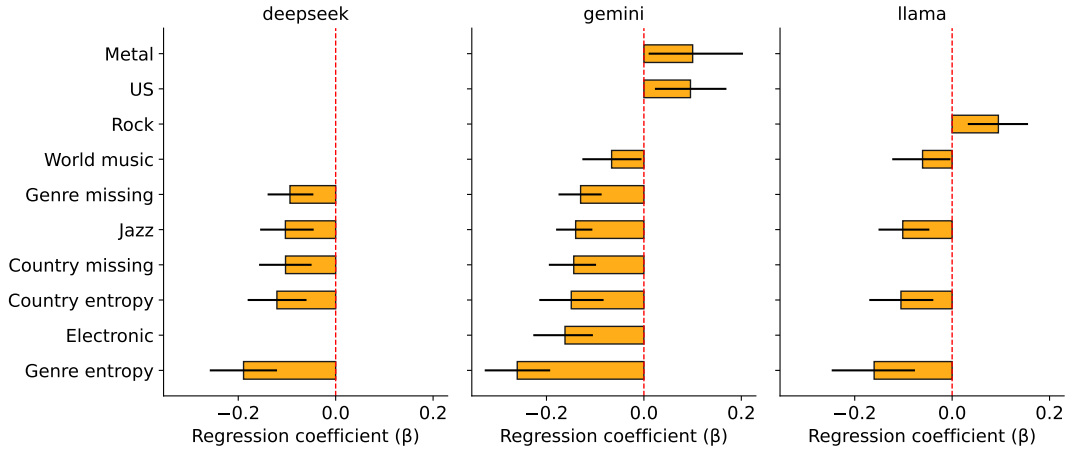

Figure 11: 5K bootstrapped estimated linear coefficients of sampled item characteristics predicting ndcg@10.

main genre: Electronic music
secondary genre: Electro pop, Nu-disco, Urban pop music, Electronic dance music (EDM), Contemporary R&B

title: Dog Music (Sleeping Music)
playcount: 8
artist: None
origin country: None
album release date: 2017
main genre: None
secondary genre: None

title: Llamado De Emergencia Deluxe
playcount: 6
artist: Daddy Yankee
origin country: Puerto Rico
album release date: 2024
main genre: World music
secondary genre: Latin electronic, Dembow, Dance pop, Latin pop music, Dancehall, Latin hip hop, Latin urban, Latin, Latin american music

title: Entra En Mi Vida
playcount: 5
artist: Sin Bandera
origin country: Mexico
album release date: 2001
main genre: Rock music
secondary genre: Latin pop music

title: Big Zen
playcount: 4
artist: None
origin country: None
album release date: 2023
main genre: None